Controlling for Confounders

The Frisch-Waugh-Lovell Theorem

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Controlling for Confounders

The Frisch-Waugh-Lovell Theorem

What does "controlling for X" actually mean?

`Causal Inference · MSc · Week 11 of 15 · 120 min`

Roadmap — the lecture

Part	Topic	Time
1 — Geometry	What regression does in space	~25 min
2 — Theorem	FWL: proof + numerical verification	~20 min
3 — Failure modes	When OLS FWL breaks	~20 min
4 — Double ML	Replace OLS with ML	~20 min
5 — Neyman	Why the "double" matters	~10 min
Checkpoints	Three exam-style questions	woven in

Welcome. Today is the algebraic engine of the entire block — Weeks 12 and 15 both build directly on what we do here, so investing in understanding FWL deeply pays compound interest later.

The goal for this lecture is unusually precise: by the end, students should be able to explain what "controlling for X" means without using the words "holding constant." That phrase is a description, not an explanation. FWL gives the actual explanation.

Pacing note: Part 1 (geometry) is the hardest section conceptually, even though it has no formulas. Slow down there. The theorem in Part 2 will land much more easily if the geometry has clicked first.

Roadmap — your payoff

Three audiences, three payoffs:

Social science — what "controlling for" *means* conceptually, not just how to do it. Parts 1–2 are your core.

CS / ML — the algorithm and precisely where it breaks. Parts 3–4 and the cross-fitting diagram.

Econ / management — when to trust an estimate and when to be suspicious. Parts 2–3 and the comparison table.

> **Note**
>
> The thread across weeks: **FWL** → **Double ML** → **CATE** (Weeks 11–12–15). Today is the algebraic engine. Everything from here builds on this.

Part 1 · Geometry

Before any formula — what is regression actually doing?

The number that should make you nervous

You run OLS of sales on ad spend: $\hat{\beta} = 3.2$

£1,000 more in ad spend → £3,200 more sales.

The problem: firms with better products advertise more *and* sell more. Quality drives both — it confounds the comparison.

> **Important**
>
> **The question for today:** when you add quality as a control, what does OLS *actually do* — and why does that fix the bias?

Think of Y, T, X as **vectors in** $\mathbb{R}^n$ — one coordinate per observation. Regression is **projection**.

> **Tip**
>
> **T is not "T with X removed."** It is the variation in T that X **cannot predict**. T is orthogonal to col(X) by construction — that is what residualising does.

Do the same for Y. Both residuals $\tilde{T}$ and $\tilde{Y}$ land in the space X knows nothing about.

We have not written a single formula yet — and we already know what $\hat{\beta}$ is.

$\hat{\beta}$ is the slope between $\tilde{Y}$ and $\tilde{T}$: both stripped of everything X can linearly explain.

"Controlling for X" means working in the space orthogonal to X.

The formula is just the algebraic statement of the picture we just drew.

> **Note**
>
> Every coefficient in every multivariate OLS you have ever run is already a partialling-out estimator. FWL proves it — and in doing so tells you exactly when the approach works and exactly when it fails.

Part 2 · The Theorem

Formalise what the geometry already showed

The Frisch-Waugh-Lovell Theorem

Theorem (FWL, 1933–1963)

The OLS coefficient $\hat{\beta}$ on T in the full regression $Y = \alpha + \beta T + \gamma' X + \varepsilon$ is *numerically identical* to the coefficient from:

Step	Operation	Result
1	Regress T on $X \to$ save residuals	$\tilde{T}$ — variation in T unexplained by X
2	Regress Y on $X \to$ save residuals	$\tilde{Y}$ — variation in Y unexplained by X
3	Regress $\tilde{Y}$ on $\tilde{T}$ (no intercept)	$\hat{\beta}$ — *identical to the full regression*

Steps 1 and 2 can be done in either order.

> ⚠ Warning
>
> **Both sides must be residualised.**
> Regressing raw Y on $\tilde{T}$ — skipping Step 2 — gives a *different* estimator.
> Raw Y still contains $\gamma' X$, the direct influence of quality on sales. Since quality correlates with $\tilde{T}$ through the unpartialled outcome side, $\hat{\beta}$ picks up part of quality's effect and inflates above the true value.
> Nothing in the regression output flags this mistake. This is the most common FWL implementation error. The diagnostic: if your FWL coefficient disagrees with the full OLS, check whether you residualised both sides.

```python
T_s = pd.Series(T, name="T");  W_s = pd.Series(W, name="W")

# Step 1 - residualise T on W
T_tilde = sm.OLS(T_s, sm.add_constant(W_s)).fit().resid

# Step 2 - residualise Y on W
Y_tilde = sm.OLS(Y, sm.add_constant(W_s)).fit().resid
```

```python
# Step 3 - regress Ỹ on T
beta_fwl  = sm.OLS(Y_tilde, T_tilde).fit().params.iloc[0]
beta_full = sm.OLS(Y, sm.add_constant(data[["T", "W"]])).fit().params["T"]

print(f"Full OLS ^  = {beta_full:.10f}")
print(f"FWL Step 3  = {beta_fwl:.10f}")
print(f"Difference  = {abs(beta_full - beta_fwl):.2e}")
```

```
Full OLS ^  = 1.5011541198
FWL Step 3  = 1.5011541198
Difference  = 2.22e-16
```

Post-lecture notebook: reproduce all three steps, then plot $\tilde{Y}$ against $\tilde{T}$.

Proof sketch — the annihilator matrix

Optional — follow the algebra or trust the picture

Define $M_X = I - X(X'X)^{-1}X'$. Three properties:

Property	Statement	Meaning
Idempotent	$M_X^2 = M_X$	Projecting twice = projecting once
Annihilates X	$M_X X = 0$	Any vector in col(X) maps to zero
Projects to col$(X)^\perp$	$M_X v \perp X$ for any v	Output is always orthogonal to X

Note $M_X T = \tilde{T}$ and $M_X Y = \tilde{Y}$ — precisely the FWL residuals. Therefore:

$$\hat{\beta}_{\text{OLS}} = (T' M_X T)^{-1} T' M_X Y = (\tilde{T}'\tilde{T})^{-1}\tilde{T}'\tilde{Y} = \hat{\beta}_{\text{FWL}} \qquad \square$$

Checkpoint 1 — the question

Part A — explain

In words: why does Step 3 of FWL give *exactly* the same coefficient as full OLS? What is it about $\tilde{T}$ that makes the equivalence hold?

Part B — diagnose

A colleague runs the three FWL steps and gets $\hat{\beta} = 1.7$. The full OLS gives $\hat{\beta} = 1.5$. Assuming no arithmetic errors — what went wrong?

. . .

Hint: what is $\tilde{T}$ orthogonal to? What does that mean for adding X to Step 3?

Checkpoint 1 — Part A

Why does Step 3 give exactly the same number as the full regression?

$\tilde{T}$ is orthogonal to $\mathrm{col}(X)$ by construction — that is literally what residualising does.

Adding X as a regressor in Step 3 would change nothing: X has no additional explanatory power for anything already orthogonal to it.

The full regression and FWL Step 3 are solving the same optimisation in the same space. Exact equality follows — not approximately, to machine precision.

> **Note**
>
> The geometry says the same thing: $\tilde{T}$ lives in the orthogonal complement of $\mathrm{col}(X)$. Regressing in that space is equivalent to the full regression restricted to that space.

Checkpoint 1 — Part B

What procedural error produces $\hat{\beta} = 1.7$ instead of 1.5?

The colleague residualised T (Step 1) correctly, but regressed **raw** Y on $\tilde{T}$ instead of $\tilde{Y}$ on $\tilde{T}$.

Raw Y still contains $\gamma' X$ — quality's direct effect on sales. Since quality correlates with $\tilde{T}$ through the unpartialled outcome side, $\hat{\beta}$ picks up part of quality's effect on sales and inflates above the true 1.5.

Rule: both variables must be residualised. Skipping either side breaks the equivalence.

> ℹ Note
>
> **Exam signal:** diagnosing a FWL / full OLS mismatch is a fair exam question. The answer always traces back to one of the two residuals being wrong or missing.

Part 3 · When It Breaks

FWL is exact — but only if you partial out the right things, correctly

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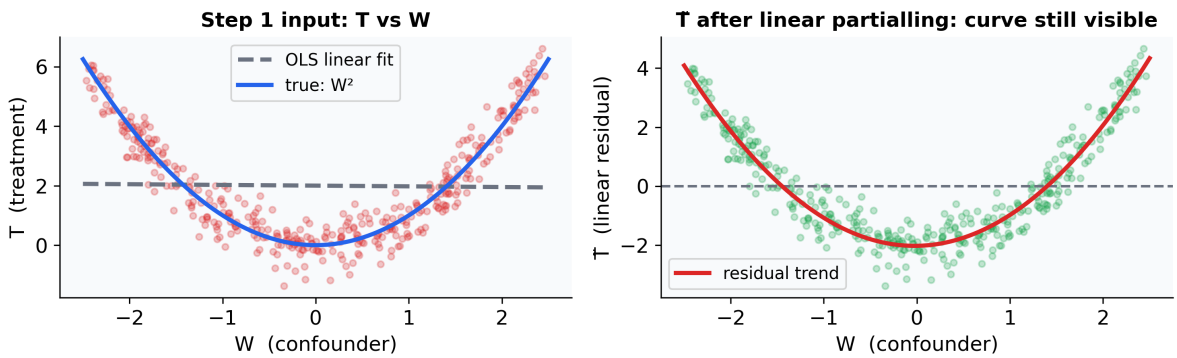
Failure mode 1 — the collider trap

FWL tells you *how* to partial out. Not *what* to partial out. That is a DAG question.

Draw your DAG before running anything. FWL on the wrong adjustment set gives an exact answer to the wrong question.

Failure mode 2 — non-linear confounders

OLS residualisation removes only the **linear** part of X's influence on T.



T is free of the *linear* effect of X — not free of X.

That is a strictly weaker guarantee.

If residuals show structure when plotted against a control, partialling is incomplete — and $\hat{\beta}$ is biased. The regression output gives no warning.

> ⚠ Warning
>
> This is the gap that Double ML closes. A flexible ML model in Steps 1 & 2 fits the true curve — not just a straight line. The residuals it produces are genuinely orthogonal to X, not just linearly orthogonal.

Failure mode 3 — high dimensions

Regime	What breaks
$p > n$	$(X'X)$ is singular — OLS cannot be computed
p close to n	OLS overfits Steps 1 & 2 — residuals are noise
Non-linear confounders	Residuals still contain confounding

Regime	What breaks
Many correlated controls	$\tilde{T}$ has near-zero variance $\rightarrow$ huge SE on $\hat{\beta}$

The right question is not "can I include more controls?" **The right question is: can I partial out X from T and Y correctly at this scale?**

OLS says no beyond a certain point. But the *structure* of FWL still holds — we just need a more flexible tool for the partialling steps. That is what Double ML provides.

Checkpoint 2 — the question

A public health researcher estimates the effect of air pollution (T) on hospital admissions (Y), controlling **linearly** for income (X).

Income has a strongly **concave** effect on both pollution exposure and health outcomes.

They report $\hat{\beta} = 0.8$ with a tight confidence interval.

Should you trust this? What exactly goes wrong — and is the bias visible in the output?

. . .

Hint: what does OLS partialling remove from T? What does it leave behind?

Checkpoint 2 — answer

No. Linear partialling removes only the linear component of income's influence.

The concave (non-linear) part remains inside $\tilde{T}$ — so $\tilde{T}$ is not truly free of income.

The bias chain:

1. OLS fits a straight line through (income, pollution) in Step 1
2. The concave residual stays in $\tilde{T}$ — high pollution at very low *and* very high incomes
3. That residual correlates with hospital admissions through income's non-linear health effect
4. Step 3 attributes that correlation to pollution

Is the bias visible in the output? No. Small standard errors and a significant p-value confirm statistical power, not correct identification. The diagnostic is to plot $\tilde{T}$ against income: a visible curve means partialling is incomplete.

Part 4 · Double Machine Learning

Replace OLS in Steps 1 & 2 — the fix is almost embarrassingly simple

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The simple insight

FWL says: partial out X from T and Y, then regress residuals.

It does not say to use OLS for the partialling.

Replace Steps 1 & 2 with any flexible ML learner:

Step	What we fit	Residual
1a	$\hat{g}(X) = \mathbb{E}[Y \mid X]$ via ML	$\tilde{Y} = Y - \hat{g}(X)$
1b	$\hat{m}(X) = \mathbb{E}[T \mid X]$ via ML	$\tilde{T} = T - \hat{m}(X)$
2	OLS of $\tilde{Y}$ on $\tilde{T}$	$\hat{\theta}$ — causal effect

Any learner works: LASSO, Random Forest, Gradient Boosting, Neural Net.

The FWL algebra is unchanged. ML does the heavy lifting in Steps 1 & 2.

Why naïve swap fails

Plugging ML directly into Steps 1 & 2 without care introduces two new problems:

Problem 1 — Regularisation

ML shrinks predictions toward the mean. Residuals are systematically too large. The signal in $\tilde{T}$ is distorted.

Problem 2 — Overfitting contamination

ML trained and evaluated on the *same* data memorises noise. In-sample residuals are too small — they under-represent true unexplained variation.

Both problems corrupt $\tilde{T}$ and $\tilde{Y}$ before Step 2 runs.

Cross-fitting fixes both.

Cross-fitting — the idea

The problem: ML memorises training data. In-sample residuals are contaminated.

The solution: for each observation, generate its residual from a model that *never saw it*.

This is K-fold cross-validation — but instead of evaluating prediction accuracy, you are generating **honest residuals** for the causal OLS.

> Tip
>
> Each fold's residuals come from a model trained on all *other* folds. No overfitting contamination. The residuals are honest — as if you had infinite separate datasets.
> K = 5 is standard: 80% training data per round, stable across runs.

Cross-fitting — the procedure

DML — the algorithm

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import KFold

kf      = KFold(n_splits=5, shuffle=True, random_state=42)
T_tilde = np.zeros(n)
Y_tilde = np.zeros(n)

for train_idx, val_idx in kf.split(X):
```

```
    for target, store in [(T, T_tilde), (Y, Y_tilde)]:
        rf = RandomForestRegressor(n_estimators=200, random_state=42)
        rf.fit(X[train_idx], target[train_idx])
        store[val_idx] = target[val_idx] - rf.predict(X[val_idx])

# Pure FWL on honest residuals
theta_dml = np.dot(T_tilde, Y_tilde) / np.dot(T_tilde, T_tilde)
```

> **Note**
>
> **Post-lecture notebook:** you implement this from scratch — same DGP, five-fold cross-fitting, and a four-estimator comparison figure reused in the Week 12 deck.

DML — when to use what

Estimator	Non-linearity	High-p	Valid SE
Naïve OLS (no controls)		—	biased
OLS + linear controls			if truly linear
OLS + FWL (manual)			exact
Double ML			**$\sqrt{n}$-consistent**

OLS + FWL is not inferior to DML in low dimensions — it is exact. DML is needed when OLS's linear partialling assumption breaks.

Use FWL to build intuition. Use DML in practice.

Part 5 · Neyman Orthogonality

Why residualising both sides is structurally necessary

Why the "double" matters

The question: DML residualises *both* T and Y. Why not just T?

Single residualisation — residualise T only

Errors in $\hat{m}(X)$ enter $\hat{\theta}$ **linearly** — proportional to δ_m. If ML converges at rate $n^{-1/4}$, bias shrinks too slowly. Confidence intervals fail at any sample size.

Double residualisation — residualise both

Errors in $\hat{m}$ and $\hat{g}$ enter as a **product** $\delta_m \times \delta_g$ — second-order small. If both converge at $n^{-1/4}$, the product is $n^{-1/2}$ — valid inference.

Why the "double" matters — the intuition

Tip

The ruler analogy
Your ruler is off by 1mm. Your scale is off by 1g.
Combined error 1mm × 1g 0. Not 2mm, not 2g.
Two small errors **multiplied** are negligible. One is not.
DML exploits this multiplicative structure — by residualising both sides, errors from the two ML models cancel each other at second order.

Neyman orthogonality — formal

Skippable — included for exam coverage

The DML score function for the partially linear model:

$$\psi(\theta;\, \hat{g}, \hat{m}) = (Y - \theta D - \hat{g}(X)) \cdot \tilde{T}$$

Neyman orthogonality means the score's derivative with respect to the nuisance functions vanishes at the truth:

$$\frac{\partial}{\partial g}\,\mathbb{E}[\psi(\theta_0;\, g, m_0)]\Big|_{g=g_0} = 0$$

Small perturbations in $\hat{g}$ and $\hat{m}$ produce zero first-order effect on $\hat{\theta}$. ML errors enter at second order only — hence the $\delta_m \times \delta_g$ product structure.

Residualising only T gives a score where $\partial\psi/\partial g \neq 0$ — error enters linearly. Residualising Y achieves orthogonality.

Exam signal: you will not be asked to derive the score. You may be asked why single residualisation fails.

Checkpoint 3 — the question

A colleague implements DML:

- Step 1: fit $\hat{m}(X) = \mathbb{E}[T \mid X]$ via Random Forest $\rightarrow \tilde{T} = T - \hat{m}(X)$
- Step 2: regress **raw** Y (not residualised) on $\tilde{T}$

They argue this saves computation and "Y doesn't need cleaning."

Are they right? What bias does this introduce — and does it vanish as $n \rightarrow \infty$?

. . .

Hint: what does raw Y still contain? How do errors in $\hat{m}$ enter the estimate — linearly or as a product?

Checkpoint 3 — answer

The colleague is wrong. Raw Y still contains $g(X)$ — X's influence on the outcome.

The score becomes $g(X) \cdot \tilde{T} + \varepsilon \cdot \tilde{T}$. The second term vanishes. The first does not — unless $\hat{m}$ is perfect, which it never is.

Setup	Bias size	Vanishes at $\sqrt{n}$ rate?
Single residualisation	$O(\delta_m)$ — linear in ML error	**No**
Double residualisation	$O(\delta_m \times \delta_g)$	**Yes**

If ML converges at $n^{-1/4}$: single $\to$ bias $\sim n^{-1/4}$ (too slow), double $\to \sim n^{-1/2}$ (valid).

This is not an asymptotic refinement. It is a structural flaw that persists at every sample size — the colleague's shortcut permanently invalidates inference.

What we covered

1. **Geometry** — "controlling for X" means working in the space orthogonal to $\mathrm{col}(X)$
2. **FWL theorem** — every OLS coefficient is a partialling-out estimator, exactly, to machine precision
3. **Three failure modes** — collider bias, non-linear confounders, high dimensions
4. **Double ML** — replace OLS partialling with ML + cross-fitting; the FWL algebra is unchanged
5. **Neyman orthogonality** — residualise both sides so errors multiply, not add

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Week 12 preview

$$\underbrace{\text{FWL}}_{\text{today}} \longrightarrow \underbrace{\text{DML}}_{\text{Week 12}} \longrightarrow \underbrace{\text{CATE}}_{\text{Week 15}}$$

Next week: implement DML end-to-end using `doubleml` and `econml` — production implementations with valid standard errors and confidence intervals.

We connect to the **R-learner** for heterogeneous treatment effects: not just the average effect, but *who benefits and by how much*.

> **Note**
>
> **Post-lecture notebook is live.** It builds directly on today — all three FWL steps in code, then breaking OLS partialling on purpose, then a from-scratch DML implementation. The figures you produce are reused in next week's deck.

Appendix — proof of M_X properties

$M_X = I - X(X'X)^{-1}X'$

Idempotent: $M_X^2 = M_X - M_X X(X'X)^{-1}X' = M_X - 0 = M_X$

Annihilates X: $M_X X = X - X(X'X)^{-1}X'X = X - X = 0$

Symmetric: $(M_X)' = I - X\big((X'X)^{-1}\big)' X' = M_X$ since $(X'X)^{-1}$ is symmetric

Therefore:

$$\hat{\beta}_{\text{OLS}} = (T'M_X T)^{-1}T'M_X Y = (\tilde{T}'\tilde{T})^{-1}\tilde{T}'\tilde{Y} = \hat{\beta}_{\text{FWL}} \qquad \square$$